Discrete Mathematics

Sets operations

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Sets - Definitions and notation

A *set* is an unordered collection of elements.

Some examples:

{1, 2, 3} is the set containing "1" and "2" and "3."
{1, 1, 2, 3, 3} = {1, 2, 3} since repetition is irrelevant.
{1, 2, 3} = {3, 2, 1} since sets are unordered.
{1, 2, 3, …} is a way we denote an infinite set (in this case, the natural numbers).
$\varnothing$ = {} is the empty set, or the set containing no elements.

Note: $\varnothing \neq \{\varnothing\}$

Representation of a Set

Sets can be represented in two ways:

1.) Roster Form: In Roster Form, the elements are inside **{}** ⇢ Curly brackets. All the elements are mentioned inside and are separated by commas. Roster form is the easiest way to represent the data in groups. For example, the set for the table of 5 will be,

A= {5, 10, 15, 20, 25, 30, 35…..}.

2.) Set-Builder Form: In the Set-builder form, elements are shown or represented in statements expressing relation among elements. The standard form for Set-builder, **A= {a: statement}**.

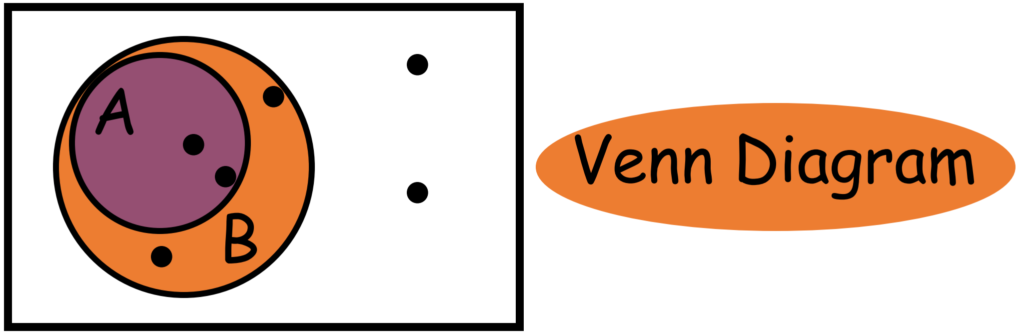
For example, $A = \{x: x = a^3, a \in N, a < 9\}$.

Sets - Definitions and notation

$x \in S$ means "x is an element of set S."
$x \notin S$ means "x is not an element of set S."

$A \subseteq B$ means "A is a subset of B."

or, "B contains A."
or, "every element of A is also in B."
or, $\forall x \, ((x \in A) \rightarrow (x \in B))$.



Sets - Definitions and notation

$A \subseteq B$ means "A is a subset of B."
$A \supseteq B$ means "A is a superset of B."

$A = B$ if and only if A and B have exactly the same elements.

iff, $A \subseteq B$ and $B \subseteq A$
iff, $A \subseteq B$ and $A \supseteq B$
iff, $\forall x\, ((x \in A) \leftrightarrow (x \in B))$.

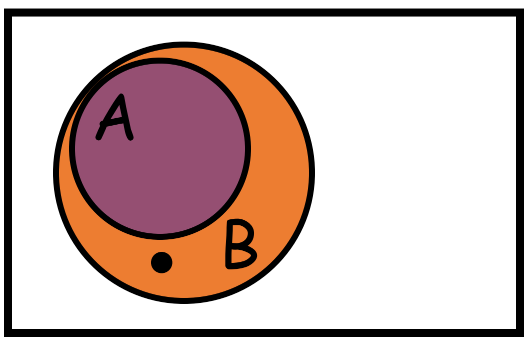
So to show equality of sets A and B, show:

- $A \subseteq B$
- $B \subseteq A$

Sets- Definitions and notation

$A \subset B$ means "A is a proper subset of B."

- $A \subseteq B$, and $A \neq B$.
- $\forall x\, ((x \in A) \rightarrow (x \in B)) \wedge \neg\forall x\, ((x \in B) \rightarrow (x \in A))$
- $\forall x\, ((x \in A) \rightarrow (x \in B)) \wedge \exists x\, \neg(\neg(x \in B) \vee (x \in A))$
- $\forall x\, ((x \in A) \rightarrow (x \in B)) \wedge \exists x\, ((x \in B) \wedge \neg(x \in A))$
- $\forall x\, ((x \in A) \rightarrow (x \in B)) \wedge \exists x\, ((x \in B) \wedge (x \notin A))$



Sets- Definitions and notation

Quick examples:

- $\{1,2,3\} \subseteq \{1,2,3,4,5\}$
- $\{1,2,3\} \subset \{1,2,3,4,5\}$

Is $\emptyset \subseteq \{1,2,3\}$?

Yes! $\forall x \, (x \in \emptyset) \rightarrow (x \in \{1,2,3\})$ holds, because $(x \in \emptyset)$ is false.

Vacuously

Is $\emptyset \in \{1,2,3\}$? No!

Is $\emptyset \subseteq \{\emptyset,1,2,3\}$? Yes!

Is $\emptyset \in \{\emptyset,1,2,3\}$? Yes!

Sets - Definitions and notation

Quiz time:

Is $\{x\} \subseteq \{x\}$? Yes

Is $\{x\} \in \{x,\{x\}\}$? Yes

Is $\{x\} \subseteq \{x,\{x\}\}$? Yes

Is $\{x\} \in \{x\}$? No

Ways to define sets

: and | are read "such that" or "where"

- Explicitly: {John, Paul, George, Ringo}
- Implicitly: {1,2,3,…}, or {2,3,5,7,11,13,17,…}
- Set builder: { x : x is prime }, { x | x is odd }. In general { x : P(x) is true }, where P(x) is some description of the set.

Ex. Let D(x,y) denote "x is divisible by y."

Give another name for

$$\{ x : \forall y\, ((y > 1) \wedge (y < x)) \rightarrow \neg D(x,y) \}.$$

Primes

Can we use **any** predicate P to define a set
$S = \{ x : P(x) \}$?

Russell's Paradox

Can we use **any** predicate P to define a set

$$S = \{ x : P(x)$$

No!

Define $S = \{ x : x \text{ is a set where } x \notin x \}$

Then, if $S \in S$, then by defn of S, $S \notin S$.

So S must not be in S, right?

But, if $S \notin S$, then by defn of S, $S \in S$.

ARRRGH!

There is a town with a barber who shaves all the people (and only the people) who don't shave themselves.

Who shaves the barber?

Cardinality

If S is finite, then the *cardinality* of S, $|S|$, is the number of distinct elements in S.

If $S = \{1,2,3\}$, $|S| = 3.$

If $S = \{3,3,3,3,3\}$, $|S| = 1.$

If $S = \varnothing$, $|S| = 0.$

If $S = \{ \varnothing, \{\varnothing\}, \{\varnothing,\{\varnothing\}\} \}$, $|S| = 3.$

If $S = \{0,1,2,3,\ldots\}$, $|S|$ is infinite. (more on this later)

Power sets

If S is a set, then the *power set* of S is
$P(s) = \{ x : x \subseteq S$

aka P(S)

We say, "P(S) is the set of all subsets of S."

If $S = \{a\}$, $P(s) = \{\varnothing, \{a\}\}$.

If $S = \{a,b\}$, $P(s) = \{\varnothing, \{a\}, \{b\}, \{a,b\}\}$.

If $S = \varnothing$, $P(s) = \{\varnothing\}$.

If $S = \{\varnothing,\{\varnothing\}\}$, $P(s)= \{\varnothing, \{\varnothing\}, \{\{\varnothing\}\}, \{\varnothing,\{\varnothing\}\}\}$.

Fact: if S is finite, $|P(s)| = 2^{|S|}$. (if $|S| = n$, $|P(s)| = 2^n$)

Cartesian Product

The *Cartesian Product* of two sets A and B is:

$$A \times B = \{ \langle a,b \rangle : a \in A \wedge b \in B\}$$

If A = {Charlie, Lucy, Linus}, and
B = {Brown, VanPelt}, then

We'll use these special sets soon!

A x B = {<Charlie, Brown>, <Lucy, Brown>, <Linus, Brown>, <Charlie, VanPelt>, <Lucy, VanPelt>, <Linus, VanPelt>}

$A_1 \times A_2 \times \ldots \times A_n = \{\langle a_1, a_2, \ldots, a_n \rangle : a_1 \in A_1, a_2 \in A_2, \ldots, a_n \in A_n\}$

A,B finite → |AxB| = ?

a) AxB
b) |A|+|B|
c) |A+B|

a) |A||B|

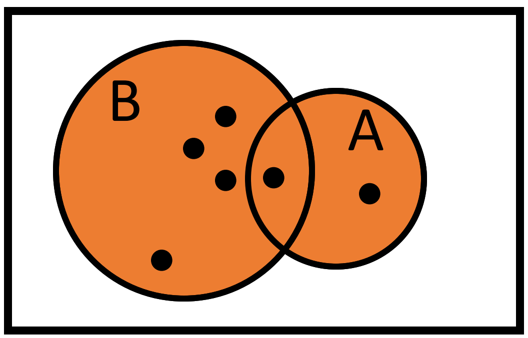
Set Theory - Operators

The *union* of two sets A and B is:

$$A \cup B = \{ x : x \in A \vee x \in B\}$$

If A = {Charlie, Lucy, Linus}, and B = {Lucy, Desi}, then

$A \cup B$ = {Charlie, Lucy, Linus, Desi}



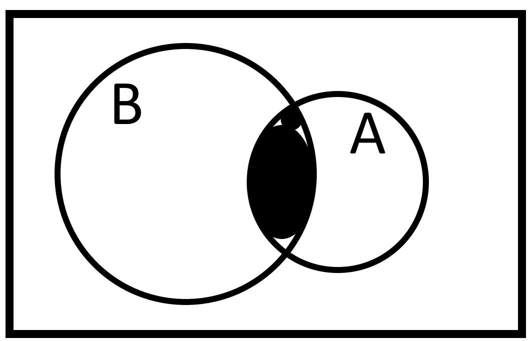
Set Theory - Operators

The *intersection* of two sets A and B is:

$$A \cap B = \{ x : x \in A \wedge x \in B\}$$

If A = {Charlie, Lucy, Linus}, and
B = {Lucy, Desi}, then

$A \cap B$ = {Lucy}



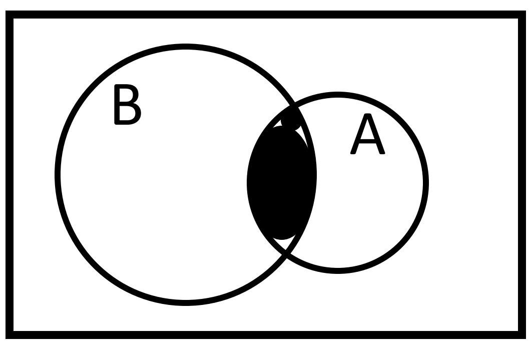
Set Theory - Operators

The *intersection* of two sets A and B is:

$$A \cap B = \{ x : x \in A \wedge x \in B\}$$

If $A = \{x : x \text{ is a US president}\}$, and
$B = \{x : x \text{ is deceased}\}$, then

$A \cap B = \{x : x \text{ is a deceased US president}\}$



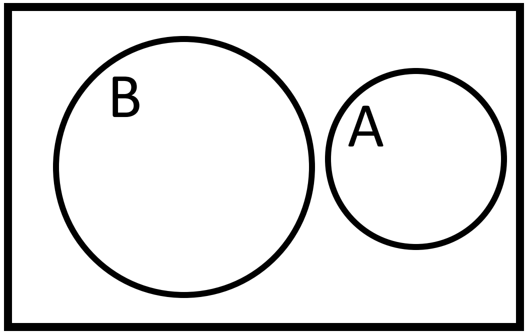
Set Theory - Operators

The *intersection* of two sets A and B is:

$$A \cap B = \{ x : x \in A \wedge x \in B\}$$

If $A = \{x : x \text{ is a US president}\}$, and
$B = \{x : x \text{ is in this room}\}$, then

$A \cap B = \{x : x \text{ is a US president in this room}\} = \varnothing$



Sets whose intersection is empty are called *disjoint sets*

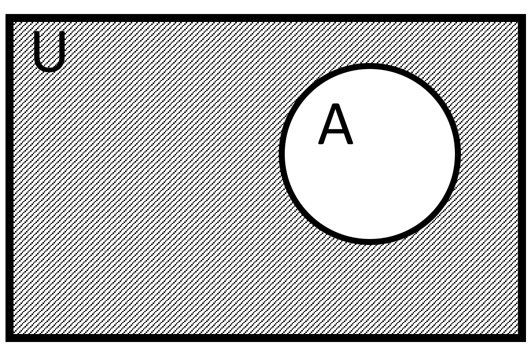
Set Theory - Operators

The *complement* ~~of~~ a set A is:

$$A = \{ x : x \notin A\}$$

If $A = \{x : x \text{ is bored}\}$, then

$\overline{A} = \{x : x \text{ is not bored}\} \;\; = \varnothing$



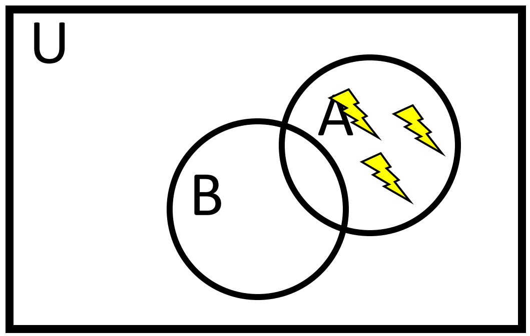
$\overline{\varnothing} = U$

and

$\overline{U} = \varnothing$

Set Theory - Operators

The *set difference*, A - B, is:



$$A - B = \{ x : x \in A \wedge x \notin B \}$$

$$A - B = A \cap \overline{B}$$

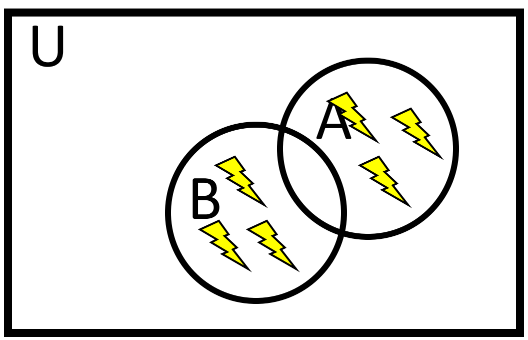
Operators

like "exclusive or"

The *symmetric difference*, $A \oplus B$, is:

$$A \oplus B = \{ x : (x \in A \wedge x \notin B) \vee (x \in B \wedge x \notin A)\}$$

$$= (A - B) \cup (B - A)$$



Set Theory - Operators

$$A \oplus B = \{ x : (x \in A \wedge x \notin B) \vee (x \in B \wedge x \notin A)\}$$

$$= (A - B) \cup (B - A)$$

Proof: $\{ x : (x \in A \wedge x \notin B) \vee (x \in B \wedge x \notin A)\}$

$$= \{ x : (x \in A - B) \vee (x \in B - A)\}$$

$$= \{ x : x \in ((A - B) \cup (B - A))\}$$

$$= (A - B) \cup (B - A)$$

Set Theory - Famous Identities

- *Identity*

$$A \cap U = A$$
$$A \cup \varnothing = A$$

- *Domination*

$$A \cup U = U$$
$$A \cap \varnothing = \varnothing$$

- *Idempotent*

$$A \cup A = A$$
$$A \cap A = A$$

Set Theory - Famous Identities

- *Excluded Middle* $A \cup \bar{A} = U$

- *Uniqueness* $A \cap \bar{A} = \emptyset$

- *Double complement* $\bar{\bar{A}} = A$

Set Theory - Famous Identities

- *Commutativity*

$$A \cup B = B \cup A$$
$$A \cap B = B \cap A$$

- *Associativity*

$$(A \cup B) \cup C = A \cup (B \cup C)$$
$$(A \cap B) \cap C = A \cap (B \cap C)$$

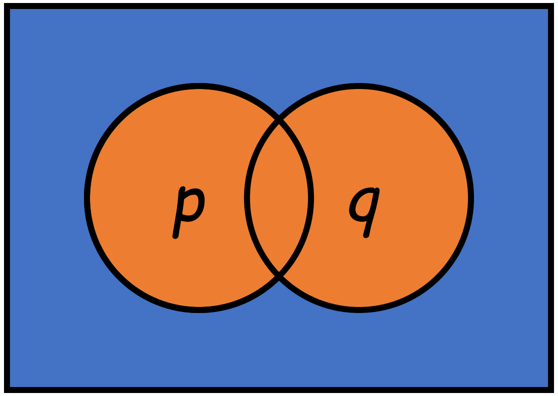
- *Distributivity*

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$$
$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

Set Theory - Famous Identities

- *DeMorgan's I* $\overline{(A \cup B)} = \overline{A} \cap \overline{B}$

- *DeMorgan's II* $\overline{(A \cap B)} = \overline{A} \cup \overline{B}$



Hand waving is good for intuition, but we aim for a more formal proof.

Set Theory - 4 Ways to prove identities

- Show that $A \subseteq B$ and that $A \supseteq B$.
- Use a membership table.
- Use previously proven identities.
- Use logical equivalences to prove equivalent set definitions.

New & important

Like truth tables

Like $\equiv$

Not hard, a little tedious

Set Theory - 4 Ways to prove identities

Prove that $\overline{(A \cup B)} = \overline{A} \cap \overline{B}$

1. $(\subseteq)$ $(x \in \overline{A \cup B}) \rightarrow (x \notin \overline{A} \cup \overline{B)} \rightarrow$ $(x \notin A \text{ and } x \notin B)$ $\rightarrow (x \in A \cap B)$

2. $(\supseteq)$ $(x \in \overline{A} \cap \overline{B}) \rightarrow (x \notin A \text{ and } x \notin B) \rightarrow$ $(x \notin A \cup B) \rightarrow (x \in \overline{A \cup B)}$

Set Theory - 4 Ways to prove identities

Prove that $\overline{(A \cup B)} = \overline{A} \cap \overline{B}$ using a membership table.

0 : x is not in the specified set

1 : otherwise

A	B	$\overline{A}$	$\overline{B}$	$\overline{A} \cap \overline{B}$	$A \cup B$	$\overline{A \cup B}$
1	1	0	0	0	1	0
1	0	0	1	0	1	0
0	1	1	0	0	1	0
0	0	1	1	1	0	1

Haven't we seen this before?

Set Theory - 4 Ways to prove identities

Prove that $\overline{(A \cup B)} = \overline{A} \cap \overline{B}$ using identities.

$$\overline{(A \cup B)} = \overline{\overline{\overline{A}} \cup \overline{\overline{B}}}$$

$$= \overline{\overline{\overline{A} \cap \overline{B}}}$$

$$= \overline{A} \cap \overline{B}$$

Set Theory - 4 Ways to prove identities

Prove that $\overline{(A \cup B)} = \overline{A} \cap \overline{B}$ using logically equivalent set definitions.

$$\overline{(A \cup B)} = \{x : \neg(x \in A \vee x \in B)\}$$

$$= \{x : \neg(x \in A) \wedge \neg(x \in B)\}$$

$$= \{x : (x \in \overline{A}) \wedge (x \in \overline{B})\}$$

$$= \overline{A} \cap \overline{B}$$

$A - B = A \cap \overline{B}$

Set Theory - A proof for us to do together.

$X \cap (Y - Z) = (X \cap Y) - (X \cap Z)$. True or False?

Prove your response.

$(X \cap Y) - (X \cap Z) = (X \cap Y) \cap (X \cap Z)'$

$= (X \cap Y) \cap (X' \cup Z')$

$= (X \cap Y \cap X') \cup (X \cap Y \cap Z')$

$= \emptyset \cup (X \cap Y \cap Z')$

$= (X \cap Y \cap Z')$

Set Theory - A proof for us to do together.

Pv that if $(A - B) \cup (B - A) = (A \cup B)$ then ______$A \cap B = \varnothing$

Suppose to the contrary, that $A \cap B \neq \varnothing$, and that $x \in A \cap B$.

a) $A \cup B = \varnothing$
b) $A = B$
c) $A \cap B = \varnothing$
d) $A-B = B-A = \varnothing$

Then x cannot be in $A-B$ and x cannot be in $B-A$.

DeMorgan's!!

Then x is not in $(A - B) \cup (B - A)$.

Do you see the contradiction yet?

But x is in $A \cup B$ since $(A \cap B) \subseteq (A \cup B)$.

Thus, $A \cap B = \varnothing$.

Applications of Sets in Computer Science

- **Data Structures:**

Set data structures store unique elements, which is useful for tasks like finding distinct values in an array or listing unique items in a playlist.

- **Database Systems:**

Set theory forms the basis of relational databases, allowing for efficient data querying and operations like joins, unions, and intersections to manipulate large datasets.

- **Algorithms:**

Sets are used in algorithms to improve performance by providing fast lookups and enabling efficient management of unique collections of elements.

Applications of Sets in Computer Science

- **Compiler Design:**

Sets are used in pattern matching and the analysis of formal languages, which is crucial for compilers to process and understand programming languages.

- **Artificial Intelligence (AI):**

Set theory provides a framework for AI systems to reason about complex problems, manipulate sets of data, and develop intelligent machines.

- **Formal Languages:**

The concepts of set theory are applied to describe and understand the structure and rules of formal languages, a foundational aspect of computer science.

Applications of Sets in Computer Science

- **Data Analysis:**

Sets help in organizing and manipulating data, making them valuable for various data analysis tasks.

- **Networking:**

Sets can be used in networking to represent different groups of nodes or connections, enabling efficient management of network resources.

- **Realtime Applications of Set:**
    - Set are used in the music player apps to list all the songs in proper order.
    - Set are used to determine the distinct values in an array.
    - Set are used to detect cycles in a Linked List.
    - Sets are used in various database operations such as performing joins.